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EDUCATION

Indian Institute of Information Technology Kalyani
Bachelor of Technology - Computer Science and Engineering

CGPA: 9.17 / 10
November 2022 – June 2026

EXPERIENCE

Resourcio Private Limited

Junior Machine Learning Developer Intern

October 2025 – January 2026

- Designed and developed backend components for FitFinder, an AI-powered product recommendation platform, enabling personalized clothing fit and style suggestions.
- Engineered data preprocessing and feature extraction pipelines to support ML-based fit prediction and user preference modeling.
- Collaborated with the ML team to develop and integrate prediction models, improving recommendation relevance and consistency across user profiles.

PROJECTS

Plant Leaf Disease Severity Estimation using Fuzzy C-Means Clustering

- Developed an end-to-end deep learning pipeline using CNNs and object detection techniques for leaf type classification and disease detection.
- Applied OpenCV-based image processing for background removal, diseased-region segmentation, and image preprocessing.
- Utilized Fuzzy C-Means Clustering to segment infected regions and quantify disease severity through pixel-level analysis and infection percentage estimation.
- Integrated a Retrieval-Augmented Generation (RAG) based chatbot using LLMs, LangChain, and vector embeddings to provide context-aware disease diagnosis, treatment recommendations, and agricultural guidance.
- Tech Stack:** Python, OpenCV, NumPy, Pandas, CNN, Fuzzy C-Means Clustering, LangChain, RAG, LLMs, Computer Vision, Image Processing.

Sehat Buddy – Medical Diagnosis QA System

- Designed and deployed a disease prediction system for Pneumonia, Kidney, and Heart diseases using hybrid Machine Learning and Deep Learning models.
- Developed an LLM-powered conversational chatbot for health-related QA using NLP and Generative AI techniques to provide context-aware responses.
- Built and integrated RESTful APIs using FastAPI to connect the frontend, ML models, and chatbot services, demonstrating strong backend development and API design skills.
- Implemented location-based doctor recommendations for post-diagnosis follow-up using geolocation and data integration.
- Tech Stack:** Python, Scikit-learn, TensorFlow, FastAPI, NLP, LLMs, Generative AI, REST APIs.

SKILLS SUMMARY

- Programming Languages:** Python, C++
- Machine Learning & Deep Learning:** Artificial Neural Networks, Convolutional Neural Networks (CNNs), and Quantitative Model Evaluation Metrics
- Generative AI:** Retrieval-Augmented Generation (RAG), LLMs, NLP
- Computer Vision:** OpenCV, Image Processing
- Frameworks and Databases :** FastAPI, Structured Query Language
- Data Analysis & Visualization:** NumPy, Pandas, Power BI

ACHIEVEMENTS

- Research paper titled “Training-Free Plant Leaf Disease Severity Estimation Using Fuzzy C-Means Clustering and Reference Palette Validation” accepted at VISAPP 2026 (Spain).